Yaroslav Zhumagulov

Contact Information

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Current Position

Postdoctoral researcher at EPFL/ Lausanne / Switzerland

supervisor Prof. Dr. Oleg Yazyev *EPFL*

2012–2016 **Bachelor studies**, *Moscow Engineering Physics Institute*, GPA 4.3/5.0
Condensed matter physics and photonics

2016–2018 **Master studies**, *Moscow Engineering Physics Institute*, GPA 4.9/5.0
Condensed matter physics and photonics

2018–2019 **Researcher**, *ITMO University*

2019–2020 **Visiting Student**, *University of Regensburg*

2020–2023 **PhD studies**, *University of Regensburg*, *magna cum laude*
Condensed matter physics and photonics

Bachelor thesis

title *Recovery of the density of states of strongly correlated systems using numerical methods*

supervisor PhD Krasavin Andrey *Moscow Engineering Physics Institute*

Master thesis

title *Investigation of magnetic, nematic and superconducting properties of high-temperature iron-based superconductors using the variational cluster approximation method*

supervisor PhD Andrey Krasavin *Moscow Engineering Physics Institute*

PhD thesis

title *Theoretical investigation of the interplay of proximity-induced spin interactions and correlated phenomena in multilayer graphene systems*

supervisor Prof. Dr. Jaroslav Fabian *University of Regensburg*

Language skills

Russian native

English fluent

German beginner

Grants, honors and awards

2016 First prize at the competition of youth scientific works. Physical Institute. P.N. Lebedev RAS and Moscow Engineering Physics Institute

2018 The best oral presentation at 5th International Youth Scientific Conference "Physics. Technologies. Innovations PTI-2018"

2015-2018 Increased scholarship for activities in science

Publications

1. Seiler, A., **Y. Zhumagulov**, Zollner, K., Yoon, C., Urbaniak, D., Geisenhof, F., Watanabe, K., Taniguchi, T., Fabian, J., Zhang, F. & Others Layer-selective spin-orbit coupling and strong correlation in bilayer graphene. *arXiv:2403.17140*. (2024)
2. **Y. Zhumagulov** & Perebeinos, V. Perspective on trions and excitons in two-dimensional atomically thin materials. *Applied Physics Letters*. **125** (2024)
3. Menushenkov, A., Ivanov, A., Neverov, V., Lukyanov, A., Krasavin, A., Yastrebtsev, A., Kovalev, Y., **Y. Zhumagulov**, Kuznetsov, A., Popov, V. & others. Direct evidence of real-space pairing in BaBiO3. *Physical Review Research*. **6**, 023307 (2024)
4. **Y. Zhumagulov**, D. Kochan, and J. Fabian, Swapping exchange and spin-orbit induced correlated phases in proximitized Bernal bilayer graphene, Physical Review B **110**, 045427 (2024)
5. **Y. Zhumagulov**, D. Kochan, and J. Fabian, Emergent correlated phases in rhombohedral trilayer graphene induced by proximity spin-orbit and exchange coupling, Physical Review Letters **132**, 186401 (2024)
6. **Y. Zhumagulov**, S. Chiavazzo, I. A. Shelykh, and O. Kyriienko, Robust polaritons in magnetic monolayers of CrI$_3$, Physical Review B **108**, L161402 (2023)
7. S. Meier, **Y. Zhumagulov**, M. Dietl, P. Parzefall, M. Kempf, J. Holler, P. Nagler, P. E. Faria Junior, J. Fabian, T. Korn, and C. Schüller, Emergent trion-phonon coupling in atomically reconstructed MoSe2−WSe2 heterobilayers, Physical Review Research **5**, L032036 (2023)
8. A. Kudlis, M. Kazemi, **Y. Zhumagulov**, H. Schrautzer, A. I. Chernov, P. F. Bessarab, I. V. Iorsh, and I. A. Shelykh, All-optical magnetization control in CrI$_3$ monolayers: A microscopic theory, Physical Review B **108**, 094421 (2023)
9. D. Kochan, A. Costa, **I. Zhumagulov**, and I. Žutić, Phenomenological Theory of the Supercurrent Diode Effect: The Lifshitz Invariant, arXiv preprint arXiv:2303.11975 (2023)
10. Z. An, P. Soubelet, **Y. Zhumagulov**, M. Zopf, A. Delhomme, C. Qian, P. E. Faria Junior, J. Fabian, X. Cao, J. Yang, A. V. Stier, F. Ding, and J. J. Finley, Strain control of exciton and trion spin-valley dynamics in monolayer transition metal dichalcogenides, Physical Review B **108**, L041404 (2023)
11. D. Gulevich, **Y. V. Zhumagulov**, V. Kozin, and I. Tokatly, Excitonic effects in time-dependent density functional theory from zeros of the density response, Physical Review B **107**, 165118 (2023)
12. **Y. V. Zhumagulov**, A. Vagov, D. R. Gulevich, and V. Perebeinos, Electrostatic and Environmental Control of the Trion Fine Structure in Transition Metal Dichalcogenide Monolayers, Nanomaterials **12**, 3728 (2022)
13. M. Kazemi, V. Shahnazaryan, **Y. Zhumagulov**, P. F. Bessarab, and I. Shelykh, Interaction of excitons with magnetic topological defects in 2D magnetic monolayers: localization and anomalous Hall effect, 2D Materials **10**, 015003 (2022)
14. **Y. Zhumagulov**, T. Frank, and J. Fabian, Edge states in proximitized graphene ribbons and flakes in a perpendicular magnetic field: Emergence of lone pseudohelical pairs and pure spin-current states, Physical Review B **105**, 205134 (2022)
15. **Y. V. Zhumagulov**, V. D. Neverov, A. E. Lukyanov, D. R. Gulevich, A. V. Krasavin, A. Vagov, and V. Perebeinos, Nonlinear spectroscopy of excitonic states in transition metal dichalcogenides, Physical Review B **105**, 115436 (2022)
16. F. S. Covre, P. Faria, V. O. Gordo, C. S. de Brito, **Y. V. Zhumagulov**, M. D. Teodoro, O. Couto, L. Misoguti, S. Pratavieira, M. B. d. Andrade, et al., Revealing the impact of strain in the optical properties of bubbles in monolayer MoSe$_2$, Nanoscale **14**, 5758 (2022)
17. A. E. Lukyanov, I. A. Kovalev, V. D. Neverov, **Y. V. Zhumagulov**, A. V. Krasavin, and D. Kochan, Strength of the Hubbard potential and its modification by breathing distortion in BaBiO$_3$, Physical Review B **105**, 045131 (2022)
18. **Y. V. Zhumagulov**, S. Chiavazzo, D. R. Gulevich, V. Perebeinos, I. A. Shelykh, and O. Kyriienko, Microscopic theory of exciton and trion polaritons in doped monolayers of transition metal dichalcogenides, npj Computational Materials **8**, 92 (2022)
19. V. D. Neverov, A. E. Lukyanov, **Y. V. Zhumagulov**, A. V. Krasavin, and V. Perebeinos, Polaronic signatures in pristine phosphorene, Physical Review Materials **5**, 054008 (2021)
20. A. E. Lukyanov, V. D. Neverov, **Y. V. Zhumagulov**, A. P. Menushenkov, A. V. Krasavin, and A. Vagov, Laser-induced ultrafast insulator-metal transition in BaBiO$_3$, Physical Review Research **2**, 043207 (2020)
21. **Y. V. Zhumagulov**, A. Vagov, D. R. Gulevich, P. E. Faria Junior, and V. Perebeinos, Trion induced photoluminescence of a doped MoS2 monolayer, The Journal of Chemical Physics **153** (2020)
22. **Y. V. Zhumagulov**, A. Vagov, N. Y. Senkevich, D. R. Gulevich, and V. Perebeinos, Three-particle states and brightening of intervalley excitons in a doped MoS 2 monolayer, Physical Review B **101**, 245433 (2020)
23. D. R. Gulevich, **Y. V. Zhumagulov**, A. Vagov, and V. Perebeinos, Nonadiabatic electron dynamics in time-dependent density functional theory at the cost of adiabatic local density approximation, Physical Review B **100**, 241109 (2019)
24. **Y. V. Zhumagulov**, A. V. Krasavin, A. Lukyanov, V. D. Neverov, A. Yaroslavtsev, and A. P. Menushenkov, Effect of Optical Excitation on the Band Structure and X-Ray Absorption Spectra of BaBiO 3-Based High-Temperature Superconductors: Ab Initio Calculation, JETP Letters **110**, 31 (2019)
25. **Y. V. Zhumagulov**, V. A. Kashurnikov, A. V. Krasavin, A. Lukyanov, and V. D. Neverov, Phase Diagram of the Two-Orbital Model for Iron-Based HTSC: Variational Cluster Approximation, JETP Letters **109**, 45 (2019)
26. A. V. Krasavin, P. V. Borisyuk, O. S. Vasiliev, **Y. V. Zhumagulov**, V. A. Kashurnikov, U. N. Kurelchuk, and Y. Y. Lebedinskii, Calculation of density of states of transition metals: From bulk sample to nanocluster, Review of scientific instruments **89** (2018)
27. V. A. Kashurnikov, A. V. Krasavin, and **Y. V. Zhumagulov**, Momentum distribution and non-Fermi-liquid behavior in low-doped two-orbital model: Finite-size cluster quantum Monte Carlo approach, Physical Review B **94**, 235145 (2016)
28. V. A. Kashurnikov, A. V. Krasavin, and **Y. V. Zhumagulov**, Electron density of states of Fe-based superconductors: Quantum trajectory Monte Carlo method, JETP letters **103**, 334 (2016)